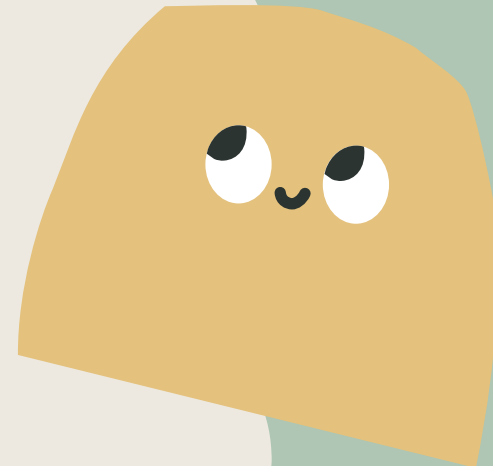
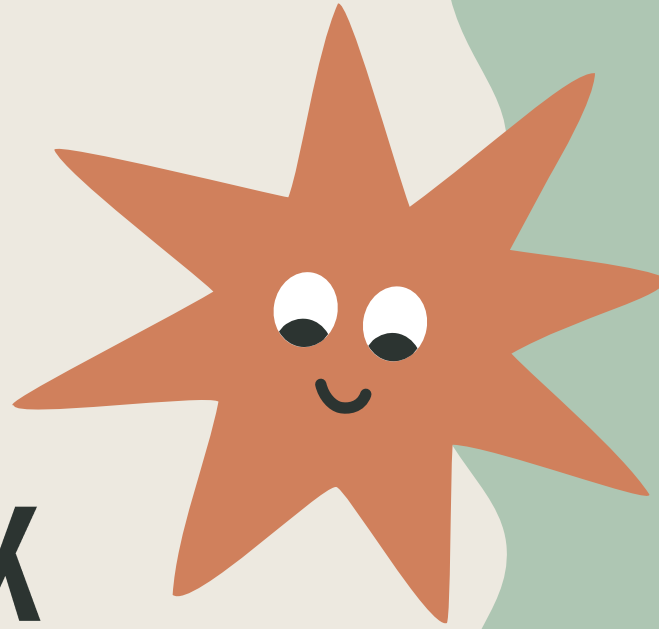


UNIT 4

DATA COMMUNICATION AND COMPUTER NETWORK

9TH COMPUTER SCIENCE
SINDH TEXTBOOK BOARD

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Agenda

Basics of Data Communication

Basic Terminologies of Data Communication

Components of a communication system

Properties of a good communication system

Transmission medium

Transmission impairments

Communication devices

Basics of computer networks

Types of computer networks

Fundamental of topologies

Standard organizations

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Basics of Data Communication

- Communication is the process of sharing a message. A conversation between two people is an example of communication.
- Data communications refers to the sharing of a virtual message between two devices. Electronic communications, like emails and instant messages and phone calls are examples of data communications.

Basic Terminologies of Data Communication

Data

The raw facts and figures are called data. The word data is derived from Latin language and it is plural of Datum. The text, numbers, symbols, images, voice and video which are also data. Data can be considered as unprocessed information.

Data Communication

Data Communication is the process of transferring data electrically from one place to another.

Data Transmission

The data transmission means emission of data in any direction via wireless or wired medium. Transmission may occur between source and destination.

Analog Signals

Analog signals are waves that change with time period and are used to represent data. An analog signal can be used to measure changes in some physical quantities such as light, sound, pressure or temperature.

Basic Terminologies of Data Communication

Digital Signals

A digital signal is an electrical signal represented as 0 or 1.

Data Rate / Bit Rate

Data rate or Bit Rate is the rate at which data is transferred. It is normally measured in bits per second. Bit is the actual binary digit which is the basic unit of data transmission. Bit can hold either 0 or 1. Data rate can be ranging from bps (bits per second) for smaller values to kbps (kilo bits per second) and mbps (megabits per second).

Signal to Noise Ratio

SNR measures how strong the signal is compared to the noise. $SNR = \text{signal strength} / \text{noise}$ – higher is better for clear communication.

Components of a communication system

Message	It is the information or data to be communicated. Common forms of information include text, numbers, pictures, audio and video.
Sender	It is the device that generates and sends a message. It can be a computer, telephone handset, etc.
Receiver	It is the device that receives a message. It can be a computer, telephone handset, etc.
Medium	It is the channel or path through which the message is carried from sender to the receiver. Some examples include twisted-pair cable, coaxial cable, radio waves, etc.
Protocol	A protocol is an agreement/rules between two parties(communication devices). Sender and receiver follow same protocols to communicate with each other. Examples include; SMTP, HTTPS, FTP, UDP etc.

Properties of a Good Communication System

Delivery

Making sure that the data is delivered is the first fundamental characteristic of any communication network. The system must be able to deliver data in correct order to the correct destination.

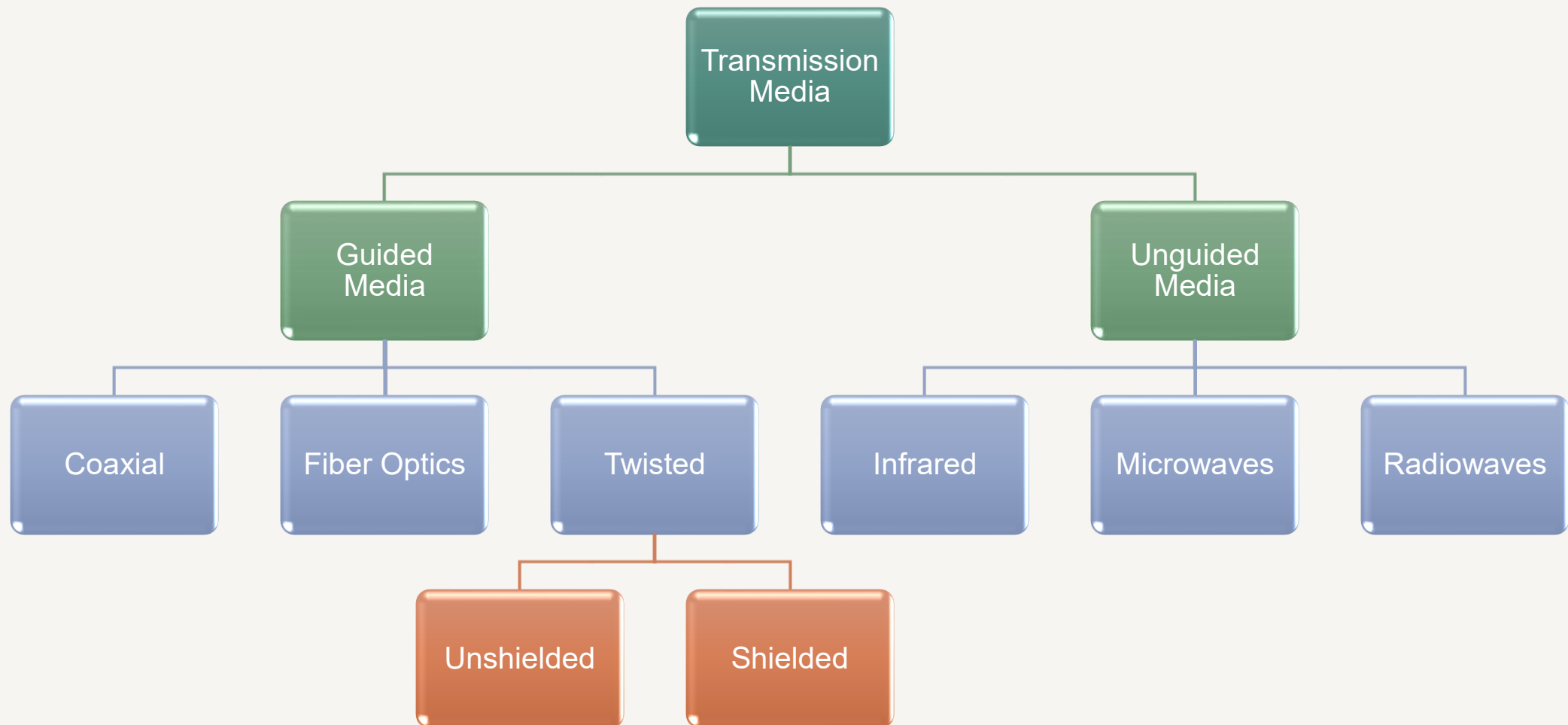
Accuracy

The system must deliver the data accurately. Data that has been altered during transmission and left uncorrected is not useful.

Timeliness

The data must be delivered in a timely manner. Late delivered data is useless.

Transmission Media



Guided Media

- In guided media signals are transmitted in a narrow pathway by using physical links.
- It is also called Wired or Bounded transmission media.
- The physical links are the cables that are tangible or have physical existence.
- There are three common types of guided media used for the networks. Each of them has its own characteristics like transmission speed, effect of noise, physical appearance, cost, etc.

Guided Media: Twisted Pair Cable

- As name suggests, this cable is made by two separate wires twisted together. A twisted pair cable is made up of insulated copper wires. The insulation and twisting of wires prevent external interference. Each pair of wires has unique color code. This type of cable is widely used in different kinds of data and voice infrastructures. There are two types of twisted pair cables:

Guided Media: Twisted Pair Cable



Unshielded Twisted Pair (UTP)

This type of cable can block interference but it is vulnerable to external interference. It is mostly used for telephonic applications. It is less expensive and easy to install.



Shielded Twisted Pair (STP)

This type of cable consists of a special coating to block external interference. It is used in fast-data-rate ethernet and also in voice and data channels of telephone lines.

Guided Media: Coaxial Cable & Fiber-Optic Cable



Coaxial Cable

Coaxial cable is also known as coax. It has an outer plastic covering containing two parallel conductors each having a separate insulated protection cover. Cable TVs and analog television networks widely use coaxial cables



Fiber-Optic Cable

In optical fiber or fiber-optic cable data is transferred in the form of light. It uses the concept of reflection of light through a core made up of glass or plastic. The core is surrounded by a less dense glass or plastic covering called the cladding. It is used for transmission of large volumes of data at very high speed.

Unguided Media

- Unguided media is also termed as wireless or unbounded transmission media.
- As the name implies, it does not require physical medium such as wire for the transmission of electromagnetic signals.
- There are three major types of Unguided Media.

Unguided Media Cont....

Radio Waves

Radio waves are also called electromagnetic waves. These are easy to generate and can penetrate through buildings. Radio waves are omnidirectional and propagated in all directions. It means that sending and receiving antennas do not need to be aligned. FM, AM radios, television and cordless phones use radio waves for transmission..

Microwaves

Microwave transmission is a line-of-sight transmission i.e. the sending and receiving antennas need to be properly aligned with each other. The distance covered by the signal is directly proportional to the height of the antenna. These are mostly used for mobile phone communications tower and television broadcast. Terrestrial and Satellite are two types of microwave transmissions. Terrestrial: Terrestrial microwaves have both stations having antennas on earth. Satellite: In satellite system, some antenna are on satellite in orbit and others are on stations on earth. They work at remote places so it can be used in mobile devices.

Infrared

It uses infrared light to transmit signals. LED is used to transmit signals and light-receivers (photodiodes) to receive signals. They use terahertz frequency. It cannot penetrate walls or other objects. Infrared light is transmitted generally line of sight (point to point). Wireless infrared communications can be used to establish short range wireless links or wireless Local Area Network.

Transmission Impairments

- Sometimes, signals traveling through transmission media lose their quality. This means that received signal is not same as the signal that was sent. This phenomenon is called transmission impairments. Transmission impairments are those defects that occur when data is transmitted. There are three causes of impairment i.e. attenuation, distortion and noise.

Transmission Impairments

Attenuation

Attenuation means loss of energy. A signal loses its energy due to the resistance of medium while it is transmitted. Its strength decreases with increase in distance. Amplifiers are used to overcome attenuation and make signal stronger again. It is measured in decibels.

Distortion

Distortion means change in the shape of the signal. A composite signal has several frequencies. When it travels through a medium different component of signal may reach at different time at destination because each component has different speed in that medium. This is called distortion. They have different phases at sender and receiver ends

Noise

Unwanted signal that mixes up with the original signal during the transmission of data is called noise. It can be induced noise, crosstalk noise, thermal noise and impulse noise which may damage the signal.

COMMUNICATION DEVICES



Switch

A switch or network switch (Figure 4.15) is a networking device that connects computers and other devices like printers, scanners and cameras on a network. Data cables from all computers and other devices of network are plugged into the switch to enable communication between them.



Router

- A Router (Figure 4.16) is a device that connects two or more networks. Routers are a combination of hardware and software. The main function of a router is to determine the optimal data path and transfer the information through that path, also known as network traffic controller.



Modem

Modem is short for Modulator and Demodulator. Modulation is the process of converting digital signals into analog signals. Demodulation is Fig: 4.15 Switch Fig: 4.16 Router 4.3 Define different communication devices Explain the function of router Differentiate among different types of modems Describe the function of switch/router Elaborate the functions of different communication devices SLOs 78 quite opposite; it converts analog signals into digital signals. Modem has the ability of sending and receiving signals that allows computers to share information with each other. This sharing of information is possible over phone lines, cables or satellite connections. .

COMMUNICATION DEVICES



Dial-up Modem

Dialup modems (Figure 4.17) use standard telephone lines to transmit and receive information. A dialup modem can be internal or external. It is important to remember that telephone lines carry only analog signals, whereas data packets sent by the computer are in digital form. In order to send these packets across a telephone line, modem converts digital signals into analog.



DSL Modem

- DSL stands for Digital Subscriber Line (Figure 4.18). Like dial-up modem DSL modem also uses telephone lines to transfer digital signals. DSL modem has a built-in network switch which enables use of twisted pair wires to deliver data and voice at high speed as compare to dial-up modem. Some DSL modems also have wireless communication functionality.



ISDN Modem

Integrated Services Digital Network (ISDN: Figure 4.19) is a digital phone connection that can transmit data, voice and video over a normal telephone line at the same time which was not done before. It is faster and expensive technology. Since ISDN work on digital transmission it converts analog voice to digital signals before transmission..



Network Interface Card (NIC)

Network cards also known as Network Interface Cards (NICs: Figure 4.20) are hardware devices that connect a computer with the network. They are installed on the mother board. They are responsible for establishing a physical connection between the network and the computer. Computer data is translated into electrical signals and sent to the network via Network Interface Cards. Modern motherboards have built-in NICs.